



Information Technology Generalist

PROG1400 – Student Workplan Grid

(Final Version)

Week / Unit	Topics / Description	Relevant Learning Outcome(s)	Assignment / Evaluation	%
Week 1 (Jan 5–9)	Course launch; OOP paradigm; Pac-Man decomposition; students select game; create Project Brief (UML v0)	LO1	Assignment – Game Concept Brief	5%
Week 2 (Jan 12–16)	Encapsulation & Abstraction; identifying objects & responsibilities	LO1, LO2	Assignment – Responsibilities Worksheet	5%
Week 3 (Jan 19–23)	UML Basics; Mermaid setup; build game Class Diagram live; students create UML Diagrams	LO1, LO2	Assignment – UML Class Diagrams	5%
Week 4 (Jan 26–30)	UML Basics; Mermaid setup; build Pac-Man Sequence Diagrams	LO1	Assignment – UML Sequence Diagrams	5%
Week 5 (Feb 2–6)	Sequence Diagrams → Code Mapping; Message passing; Interaction modeling (WS05)	LO2	Workshop 05 – From State Diagrams to Python Code (Formative)	2.5%
Week 6 (Feb 9–13)	State Machines & Behavior Modeling; Implementing FSM logic in classes (WS06)	LO1, LO2	Workshop 06 – State Implementation (Formative)	2.5%
Week 7 (Feb 16–20)	Inheritance & Polymorphism; Base classes; <code>super()</code> ; overriding methods (WS07)	LO2	Workshop 07 – Inheritance Implementation (Formative)	2.5%
Week 8 (Feb 23–27)	Engine Architecture & Separation of Concerns; Game loop structure; Modular design (WS08)	LO2, LO3	Workshop 08 – Engine Refactor (Formative)	2.5%

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Week 9 (Mar 2–6)	Managing Collections & Object Lifecycle; Lists of entities; Safe iteration patterns (WS09)	LO3	Workshop 09 – Dynamic Object Management (Formative)	2.5%
Week 10 (Mar 9–13)	Debugging & Refactoring Fundamentals; Code smells; Trace tables; Step debugging; Clean code improvements (WS10)	LO3, LO4	Workshop 10 – Debug & Refactor Lab (Formative)	2.5%
Week 11 (Mar 16–20)	Midterm Practical Evaluation; Code comprehension; UML interpretation; Debug exercise; Design explanation	LO1, LO2, LO3	Midterm Practical Evaluation (Graded)	25%
Week 12 (Mar 23–27)	Break Week – No Classes	—	—	—
Week 13 (Mar 30–Apr 3)	Introduction to Pygame; Window creation; Event loop; Rendering OOP objects visually (WS11)	LO2	Workshop 11 – Playable Window (Formative)	2.5%
Week 14 (Apr 6–10)	Real-time movement; Collision detection; Score integration; Game state management in Pygame (WS12)	LO2, LO3	Workshop 12 – Playable Prototype (Formative)	2.5%
Week 15 (Apr 13–15)	Refactoring, Optimization & Enhancements; AI tuning; Feature expansion; Portfolio polish (WS13)	LO3, LO4	Final Game Project + Demo + Reflection (Graded)	35%